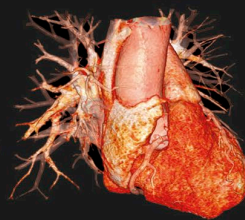
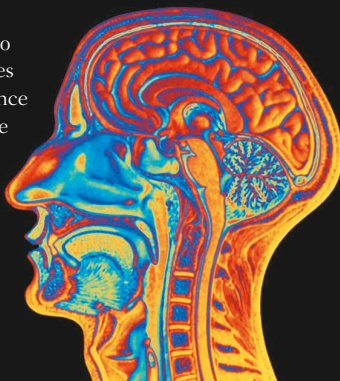


MRI AND CT SCANNING

In computerized tomography (CT) an X-ray scanner is used with a computer to build up cross-sectional images of tissues of different density. In magnetic resonance imaging (MRI), magnets are used to line up atoms in the body, then radio waves throw the atoms out of alignment. As they realign, the atoms emit signals that are used to create an image.

MRI SCAN OF HEAD

A coloured MRI scan of the mid-line of the head in side view; visible structures including the brain and spinal cord, the nasal cavity, and the tongue.



CT SCAN OF THE HEART

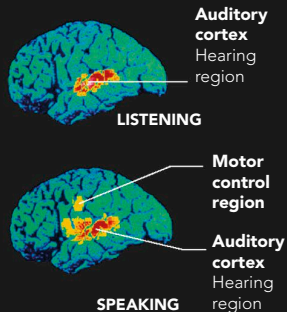
A 3-D CT scan of the heart from the right side; showing the large aorta (main artery, centre top) and some of the blood-vessels of the lungs.

NUCLEAR MEDICINE IMAGING

In nuclear medicine imaging, a radioactive substance (radionuclide) is injected and absorbed by the area to be imaged. As the substance decays it emits gamma rays, which a computer forms into an image. This type of imaging can help to diagnose disorders such as cancers and heart diseases. Nuclear medicine imaging scans like positron emission tomography (PET) and single-photon emission computed tomography (SPECT) give data about the function of a tissue rather than detailed anatomy.

PET SCAN

PET scans show function rather than anatomy. These images reveal the brain's activity as the subject listened to spoken words and then both listened to and repeated the words.



ULTRASOUND

High-frequency sound waves emitted by a device called a transducer pass into the body and echo back as electrical signals. A computer processes the signals to create images.

FETAL ULTRASOUND

Ultrasound is a very safe technique, commonly used to monitor fetal development in the uterus.

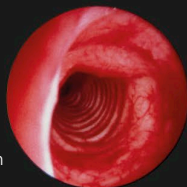


ENDOSCOPY

Endoscopes are flexible or rigid tubes inserted into the body to view its interior, perform surgical procedures, or both. They carry a light source and instruments may be passed down them.

TRACHEA

An endoscopic view of the trachea (windpipe) shows the hoops of cartilage that maintain its shape.



ELECTRICAL ACTIVITY

Sensor pads applied to the skin detect electrical activity in muscles and nerves. The signals are displayed as a trace line. This technique includes electrocardiography (ECG) of the heart (see below).

